

Documentation

A guide to using L^AT_EX File Template v.3.0.0.

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July 2026

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1 Introduction

Thank you for choosing to use L^AT_EX File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on my GitHub: <https://github.com/MatthewEmer>.

This document is intended for academics (with special focus on those in the fields of computer science or mathematics) who want to write papers or other related documents in a consistent format which ties in with standard practises from other published papers. This guide consists of setup instructions to download the files and create your first document, before moving onto the custom commands/environments designed to speed up academic writing. Before using this guide, I would suggest that you have some practise with some basic L^AT_EX commands using a guide such as <https://www.overleaf.com/learn>.

This guide is accurate as of Version 3.0.0.

2 Document Setup

2.1 Project Files

The actual L^AT_EX file template is contained in the folder *The Template* and consists of four files: *template.tex*, *_documentContent.tex*, *_appendix.tex*, and *template.pdf*. To use the template, all four files must be downloaded and placed in the same folder.

2.1.1 _appendix.tex

This file allows you to add an optional appendix to the end of your document which is not counted towards your page count. By default it contains a unnumbered part title which has been inserted into the table of contents.

2.1.2 _documentContent.tex

This file allows you to add all the main content of the document you are trying to write. It will be inserted after the table of contents, but before the appendix (if you choose to add one). By default it contains an example section and definition. Only pages in this file are counted towards the page count of the document.

2.1.3 template.pdf

This file contains the output of the L^AT_EX compiler. If you want to rename this document while still working on it, then delete this version, alongside any build files your compiler has created and change the name of *template.tex*.

2.1.4 template.tex

This file sets up the structure of the document, as well as acting as a compiler for the other two *.tex* files, ensuring they get inserted into the right sections of the document. When creating your document, you need to edit lines 12-15 of this file to create the title page, alongside the document header and footer, as described in Table 1.

Line	Variable Name	Description
12	thetitle	The document's title. This will be placed in the header.
13	thesubtitle	The document's (optional) subtitle. This will be placed in the header.
14	theauthor	The list of authors. This will be placed in the footer.
15	thedata	Replace the code if you want a static date rather than date of last compile.

Table 1: Lines to edit.

2.2 Optional Files

2.2.1 Storing Images

If you choose to include figures in your project, then you will need to create a subfolder labelled *Images* where you can place the image files for the project.

2.2.2 GitHub Repositories

If you are placing the document within a GitHub repository, you may want to copy over the *.gitignore* from this repository as it is set up to ignore any L^AT_EX build files.

3 Custom Commands

3.1 General Tools

3.1.1 Element Spacing

Command 3.1.1.1: `\separate`

The `separate` command inserts vertical spacing between two elements whilst simultaneously removing the new paragraph indent from the following element. You can specify the amount of spacing as described in Table 2 and as seen in Example 3.1.1.1.

No.	Name	Description	Example Value
1	Spacing	The number of points of vertical spacing.	5

Table 2: Parameters for `separate`.

Example 3.1.1.1: `\separate{3}` produces the separation as seen between the Command line and the description of `separate`.

3.1.2 Text Formatting

Command 3.1.2.1: `\colouredText`

The `colouredText` command takes the given text and outputs it in a different colour, as chosen by the writer. This can be seen in Example 3.1.2.1 and as described in Table 3.

No.	Name	Description	Example Value
1	Colour	The colour to change the text to.	blue
2	Text	The text you want outputted in colour.	Hello World

Table 3: Parameters for `colouredText`.

Example 3.1.2.1: `\colouredText{blue}{Hello World}` produces **Hello World**.

Command 3.1.2.2: `\comment`

The `comment` command inserts notes into the document in red so they can be found more easily. This can be seen in Example 3.1.2.2 and as described in Table 4.

No.	Name	Description	Example Value
1	Text	The text you want outputted in colour.	Hello World

Table 4: Parameters for `comment`.

Example 3.1.2.2: `\comment{Hello World}` produces **Hello World**.

3.1.3 Date Formatting

Command 3.1.3.1: `\yeardate`

The `yeardate` command takes an adjacent date variable and reformats it into the four-digit year before outputting to the page as seen in Example 3.1.3.1.

No.	Name	Description	Example Value
	n/a		

Table 5: Parameters for `yeardate`.

Example 3.1.3.1: `\date{\yeardate\today}` produces 2026.

Command 3.1.3.2: `\monthyeardate`

The `monthyeardate` command takes an adjacent date variable and reformats it into the full month and four-digit year before outputting to the page as seen in Example 3.1.3.2.

No.	Name	Description	Example Value
	n/a		

Table 6: Parameters for `monthyeardate`.

Example 3.1.3.2: `\date{\monthyeardate\today}` produces July 2026.

3.2 Tables and Figures

3.2.1 Inserting Tables

Command 3.2.1.1: `\inserttable`

The `inserttable` command simplifies the process of inserting tables into your document as described in Table 7 and seen in Example 3.2.1.1.

No.	Name	Description	Example Value
1	Columns	The required columns, separated by s.	<code>c c c c</code>
2	Caption	The caption to appear under the table.	Parameters for inserting tables.
3	Headers	The headers, separated by ampersands.	No. & Name & ...
4	Content	The content to be placed in the table.	

Table 7: Parameters for `inserttable`.

Example 3.2.1.1: `\inserttable{c|c|c|c}{Parameters...}{No. & Name & ...}{1 & Columns & ...}` produces Table 7.

3.2.2 Inserting Figures

Command 3.2.2.1: `\insertimage`

The `insertimage` command simplifies the process of inserting figures into your document as described in Table 8 and seen in Example 3.2.2.1.

No.	Name	Description	Example Value
1	File name	The file name of the image (must be in <i>Images</i> .)	durham.jpg
2	Figure caption	The caption to appear under the figure.	Durham Cathedral.
3	Width	The amount of the page width the image should occupy.	0.3

Table 8: Parameters for `insertimage`.

Example 3.2.2.1: `\insertimage{durham.jpg}{Durham Cathedral}{0.3}` produces Figure 1.

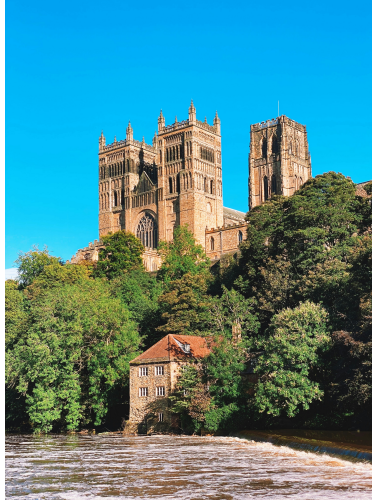


Figure 1: Durham Cathedral

3.2.3 Referencing Tables and Figures

Command 3.2.3.1: `\elemref`

The `elemref` command allows you to easily reference tables and figures with clickable links. This is described in Table 9 and as seen in Example 3.2.3.1.

No.	Name	Description	Example Value
1	Label	The label of the element being linked to.	tab:Parameters for elemref.
2	Type	The type of element being linked to.	Table

Table 9: Parameters for `elemref`.

Example 3.2.3.1: `\elemref{tab:Parameters for elemref.}{Table}` produces the reference Table 9.

3.3 Academic Environments

3.3.1 Basic Environments

Command 3.3.1.1: `\definition`

The `definition` command creates an environment for setting out a definition with automatic numbering with inbuilt labels so that you can reference the definition later in the document. Its parameters can be seen in Table 10 and an example of its use can be seen in Example 3.3.1.1.

No.	Name	Description	Example Value
1	Label	The word being defined so it can be easily referenced.	example
2	Definition	The actual definition of the word (with the word underlined).	An <u>example</u> is...

Table 10: Parameters for `definition`.

Example 3.3.1.1: `\definition{example}{An \underline{example} is what this is.}` produces Definition 3.3.1.1

Definition 3.3.1.1: An example is what this is.

Command 3.3.1.2: `\conjecture`

The conjecture command creates an environment for setting out a conjecture with automatic numbering with inbuilt labels so that you can reference the conjecture later in the document. Its parameters can be seen in Table 11 and an example of its use can be seen in Example 3.3.1.2.

No.	Name	Description	Example Value
1	Label	The conjecture's name.	Someone's Conjecture
2	Definition	The content for the conjecture.	Someone's Conjecture states...

Table 11: Parameters for conjecture.

Example 3.3.1.2: `\conjecture{Someone's Conjecture}{Someone's Conjecture states something or other.}` produces Conjecture 3.3.1.1

Conjecture 3.3.1.1: Someone's Conjecture states something or other.

3.3.2 Environments with Proofs or Solutions

Add theorem, lemma, proposition, and standard problem here.

3.3.3 Sub-Environments

Add proof and solution here.

3.3.4 Customisable Environments

Add numberedentry here.

3.3.5 Referencing Environments**Command 3.3.5.1:** `\preenvref`

The preenvref command allows you to easily reference academic environments with clickable links before they are defined in the document. This is described in Table 12 and as seen in Example 3.3.5.1.

No.	Name	Description	Example Value
1	Label	The label of the environment being linked to.	ex:cmd:postenvref
2	Type	The type of environment being linked to.	Example
3	Counter	The name of the counter for that environment.	examplenumber

Table 12: Parameters for preenvref and postenvref.

Example 3.3.5.1: `\preenvref{ex:cmd:preenvref}{Example}{examplenumber}` produces the reference Example 3.3.5.1.

Command 3.3.5.2: `\postenvref`

The postenvref command allows you to easily reference academic environments with clickable links after they are defined in the document. This is described in Table 12 and as seen in Example 3.3.5.2.

Example 3.3.5.2: `\postenvref{ex:cmd:preenvref}{Example}{examplenumber}` produces the reference Example 3.3.5.1.

3.4 Mathematics**3.4.1 Basic Mathematics Notation**

Add set notation here.

3.4.2 Vectors and Matrices

Add dot product, hessian, lagrange, and jacobian here.

3.5 Computer Science

3.5.1 Set Notation

Add joins here.

3.5.2 Algorithms

Add insertalgorithm here.